

Between Models:
Counterfactual Frontiers across Causal-Response Experiments
Online Appendix

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Abstract

This appendix contains proofs, inference details, simulation design, and additional calculations for the monetary application. The notation and target definitions follow the main text.

Appendix A. Model identification and factorization

A.1. Conventions

All response codomains are finite-dimensional Euclidean spaces with their usual topology. The model envelope is a disjoint union, so a model instance retains its family label. For a map $S : X \rightarrow Y$, its fiber at s is $S^{-1}(s)$. A factor map has domain $S(X)$. Each family keeps its own parameterization and manifold chart.

LEMMA A1 (Closure of a singleton image). *If a set $A \subseteq \mathbb{R}^q$ is empty, then $\text{cl } A$ is empty. If every element of a nonempty A equals a , then $\text{cl } A = \{a\}$.*

PROOF. The empty set is closed. A singleton in a Euclidean space is closed, and a nonempty set whose elements all equal a is the singleton $\{a\}$. $\square$

A.2. Proof of identification as factorization

PROOF OF THEOREM 2. Suppose first that T_p is constant on every fiber of S_A . For $s \in S_A(\mathfrak{E})$, choose any $M_s \in \mathfrak{E}$ satisfying $S_A(M_s) = s$ and define

$$\lambda_p(s) = T_p(M_s).$$

If M'_s is another preimage of s , fiber constancy gives $T_p(M'_s) = T_p(M_s)$, so the definition is independent of the choice. For every $M \in \mathfrak{E}$, $\lambda_p\{S_A(M)\} = T_p(M)$, which proves factorization.

If $\tilde{\lambda}_p$ is another map satisfying $T_p = \tilde{\lambda}_p \circ S_A$, then for any $s \in S_A(\mathfrak{E})$ and any M_s in its preimage,

$$\tilde{\lambda}_p(s) = \tilde{\lambda}_p\{S_A(M_s)\} = T_p(M_s) = \lambda_p(s).$$

Thus the factor is unique on the empirical image.

For the reverse implication, if $T_p = \lambda_p \circ S_A$ and $S_A(M) = S_A(M')$, then $T_p(M) = \lambda_p\{S_A(M)\} = \lambda_p\{S_A(M')\} = T_p(M')$. For a nonempty fiber, its raw policy image is therefore a singleton, and Lemma A1 proves the identified-set conclusion. A response outside $S_A(\mathfrak{E})$ has an empty fiber and receives coverage-failure status. $\square$

A.3. Differential factorization

PROOF OF THEOREM 3. Assume $\ker D_f \subseteq \ker C_{f,p}$. For $y \in \text{Im } D_f$, choose h such that $D_f h = y$ and define

$$\Lambda_{f,p,\theta_0}(y) = C_{f,p}h.$$

If $D_f \tilde{h} = y$, then $h - \tilde{h} \in \ker D_f$. Kernel inclusion implies $C_{f,p}(h - \tilde{h}) = 0$, hence $C_{f,p}h = C_{f,p}\tilde{h}$; the definition is well defined.

For $y_i = D_f h_i$ and scalars a_i ,

$$\Lambda_{f,p,\theta_0}(a_1 y_1 + a_2 y_2) = \Lambda_{f,p,\theta_0}\{D_f(a_1 h_1 + a_2 h_2)\} = C_{f,p}(a_1 h_1 + a_2 h_2),$$

so Λ_{f,p,θ_0} is linear. By construction, $\Lambda_{f,p,\theta_0} D_f h = C_{f,p} h$ for every h . If $\tilde{\Lambda} D_f = C_{f,p}$, then for every $y = D_f h$, $\tilde{\Lambda} y = C_{f,p} h = \Lambda_{f,p,\theta_0} y$; uniqueness holds on $\text{Im } D_f$.

For the reverse implication, suppose $C_{f,p} = \Lambda D_f$. If $h \in \ker D_f$, linearity gives $C_{f,p} h = \Lambda(0) = 0$, so $h \in \ker C_{f,p}$. $\square$

PROOF OF THEOREM 4. The constant-rank theorem provides neighborhoods U_1 of θ_0 and V_1 of $S_{A,f}(\theta_0)$, and continuously differentiable coordinate changes under which

$$S_{A,f}(x, z) = (x, 0),$$

where $x \in \mathbb{R}^{r_f}$ and z indexes the remaining coordinates. Shrink U_1 to a product neighborhood $U_0 = X_0 \times Z_0$ whose vertical slices Z_0 are connected. The tangent vectors $(0, v)$ span $\ker DS_{A,f}(x, z)$. By the kernel inclusion,

$$D_z T_{p,f}(x, z) v = 0 \quad \text{for every } v.$$

Thus $D_z T_{p,f}(x, z) = 0$. For fixed x and any $z_0, z_1 \in Z_0$, integrate the derivative along a piecewise continuously differentiable path in the connected coordinate ball Z_0 to obtain $T_{p,f}(x, z_0) = T_{p,f}(x, z_1)$. Define $\lambda_{f,p}(x, 0) = T_{p,f}(x, z_0)$ for any $z_0 \in Z_0$. This is well defined and continuously differentiable because it can be represented using a fixed smooth local section, for example $z = 0$. Composition gives (32).

For uniqueness, any factor map on $S_{A,f}(U_0)$ must take the value $T_{p,f}(x, z)$ at $(x, 0)$, so two such maps coincide. $\square$

A.4. Cross-family agreement

PROOF OF COROLLARY 2. Suppose the overlap condition holds. If two models $M_f(\theta)$ and $M_g(\vartheta)$ have the same training response s , then

$$T_{p,f}(\theta) = \lambda_{f,p}(s) = \lambda_{g,p}(s) = T_{p,g}(\vartheta).$$

The same argument covers $f = g$. Hence T_p is constant on every envelope fiber and Theorem 2 applies.

For the reverse implication, if the envelope identifies the target, take any s in a pairwise overlap and choose preimages θ and ϑ . Fiber constancy gives $\lambda_{f,p}(s) = T_{p,f}(\theta) = T_{p,g}(\vartheta) = \lambda_{g,p}(s)$. $\square$

A.5. The three status conditions

PROOF OF PROPOSITION 1. Write $F = \mathfrak{F}_A(s_A^\dagger; \mathfrak{C})$. Coverage failure is $F = \emptyset$, whereas conditional policy-identification failure includes $F \neq \emptyset$, so the two events are mutually exclusive. If $F = \emptyset$, its held-out image is empty and held-out relation failure follows. For a nonempty example with policy-identification and held-out relation failure, take $F = \{M_1\}$ and $s_B^\dagger \neq S_B(M_1)$. For a nonempty example with policy-identification failure and held-out relation failure, take $F = \{M_1, M_2\}$ with $T_p(M_1) \neq T_p(M_2)$ and choose $s_B^\dagger$ outside $\{S_B(M_1), S_B(M_2)\}$. These constructions establish the stated compatibility relations. $\square$

Appendix B. Testing-risk fibers, certification, and approximation

B.1. Local response metric

PROOF OF PROPOSITION 2. Continuous differentiability gives $S_{A,f}(\theta_f^0 + h/\sqrt{T}) - S_{A,f}(\theta_f^0) = T^{-1/2}D_f h + o(T^{-1/2})$ uniformly for h in a compact set. Substitution of $\Sigma_{A,T} = \Omega_{A,f}/T$ gives (10). For $\theta = \varphi(\eta)$, the chain rule gives $D_{f,\eta} = D_{f,\theta}D\varphi(\eta_0)$ and the congruence in (11). Finally, $h'H_f h = (D_f h)' \Omega_{A,f}^{-1}(D_f h)$, so positive definiteness gives $\ker H_f = \ker D_f$. $\square$

B.2. Testing affinity

For a deterministic test with rejection region A , the sum of conditional errors is $P_0(A) + P_1(A^c) = 1 - \{P_1(A) - P_0(A)\}$. Taking the infimum over randomized tests gives $1 - \sup_A |P_1(A) - P_0(A)|$, which proves the identity in (12).

PROOF OF PROPOSITION 3. Let the Gaussian means be μ_0 and μ_1 , whiten their difference, and write $\delta = \Sigma^{-1/2}(\mu_1 - \mu_0)$ with $\|\delta\| = c$. Projection on $e = \delta/c$ produces unit variance and a mean difference c . The likelihood-ratio threshold is the midpoint, so each conditional error equals $\Phi(-c/2)$. Their sum is $2\Phi(-c/2)$, and division by two gives the Hansen-Sargent detection-error probability. $\square$

B.3. Coverage and sign separation

PROOF OF PROPOSITION 4. The inclusion $\mathfrak{F}_A(s_A^\dagger; c_1) \subseteq \mathfrak{F}_A(s_A^\dagger; c_2)$ for $c_1 \leq c_2$ makes every $U_j(c)$ weakly increasing. Compactness and continuity make each constrained infimum in (16) attainable whenever its feasible set is nonempty. If $c < c_{\text{sign}}$, a compatible point with $T_{j,f}(\theta) \geq 0$ would give $\delta_{f,j}^{\text{sign}} \leq c$, a contradiction. If $c \geq c_{\text{sign}}$, an attaining sign-reversal point belongs to the radius- c fiber. The definition of c_{entry} supplies a nonempty component in every family for $c \geq c_{\text{entry}}$. $\square$

PROOF OF COROLLARY 1. For $\rho \in [0, 1]$, every weight ρ^h is nonnegative and the weight at $h = 0$ equals one. A finite weighted sum of negative coordinates is negative. $\square$

PROOF OF PROPOSITION 5. For each ξ , Proposition 4 gives the common-negative-sign interval $[c_{\text{entry}}(\xi), c_{\text{sign}}(\xi))$. This interval is nonempty exactly when $\Delta(\xi) > 0$. If $\underline{\Delta}(\Xi) > 0$, then $\Delta(\xi) > 0$ for every $\xi \in \Xi$. If some ξ_0 has $\Delta(\xi_0) \leq 0$, its interval is empty, so the collection criterion fails. $\square$

B.4. Continuous certificates

PROOF OF PROPOSITION 6. Write $x_i = \Sigma_{A,T}^{-1/2} \{S_{A,f}(\theta_{f,i}) - s_A^\dagger\}$. Convexity of the norm gives $\|\sum_i \lambda_i x_i\| \leq \sum_i \lambda_i \|x_i\| \leq c$. Each scalar policy coordinate is affine in the same simplex weights, so its extrema over the simplex occur at vertices. Equation (23) and the margin conclusion follow. $\square$

PROOF OF PROPOSITION 7. For any $\theta \in \mathcal{N}_f(s_A^\dagger; c)$, choose $\theta_g \in \mathcal{G}_f(c)$ with $\|\theta - \theta_g\| \leq \varepsilon_f$. The Lipschitz and numerical bounds give $T_{j,f}(\theta) \leq \widehat{T}_{j,f}(\theta_g) + L_{f,j}\varepsilon_f + e_{f,j}$. Taking the supremum over θ and the maximum over families gives (24). $\square$

A boxwise interval implementation can replace the covering calculation. For each parameter box B , it verifies either a distance lower bound above c or a target upper bound below zero. The calculation records closed domains, equilibrium checks, parameter-box widths, response truncation bounds, and solver allowances.

B.5. Envelope monotonicity

PROOF OF LEMMA 1. Define the invertible matrix L and multiply to obtain

$$L = \begin{pmatrix} I & 0 \\ -\Sigma_{KJ}\Sigma_{JJ}^{-1} & I \end{pmatrix}, \quad L\Sigma L' = \text{diag}(\Sigma_{JJ}, \Sigma_{K|J}).$$

Invertibility gives $r'\Sigma^{-1}r = (Lr)'(L\Sigma L')^{-1}(Lr)$, which is (21). The second quadratic form is nonnegative. Hence each model's distance weakly increases when K is added. The distance sublevel sets are nested, and taking the infimum over the same sign-reversal set proves frontier monotonicity, including the extended-real empty-set convention. $\square$

PROOF OF THEOREM 1. Envelope inclusion gives $\mathfrak{F}_A(s; \mathfrak{E}_1) \subseteq \mathfrak{F}_A(s; \mathfrak{E}_2)$. Applying T_p preserves inclusion of raw policy images, and closure is monotone, which yields (25). $\square$

B.6. Approximate factorization

PROOF OF LEMMA 2. Take M and M' in the raw set defining (35). Adding and subtracting the fitted relation values gives

$$\begin{aligned} \|T_p(M) - T_p(M')\|_{W_p} &\leq \|T_p(M) - \lambda_p\{S_A(M)\}\|_{W_p} + \|\lambda_p\{S_A(M)\} - \lambda_p\{S_A(M')\}\|_{W_p} \\ &\quad + \|\lambda_p\{S_A(M')\} - T_p(M')\|_{W_p}. \end{aligned}$$

The first and third terms are at most ϵ_C . The middle term is at most $2L_p\epsilon_A$ by the triangle inequality in response space. Every pair in the raw image is therefore separated by at most $2(\epsilon_C + L_p\epsilon_A)$, and continuity of finite-dimensional norms carries the bound to the closure. $\square$

B.7. Visibility strength

Suppose $C_{f,p} \neq 0$. The affine set $\{h : C_{f,p}h = 1\}$ is nonempty. Under $\ker D_f \subseteq \ker C_{f,p}$, Theorem 3 gives $C_{f,p} = \Lambda D_f$. A sequence in this affine set with $D_f h_n \rightarrow 0$ would imply $1 = \Lambda D_f h_n \rightarrow 0$. Finite-dimensional norm equivalence and positive definiteness of Σ_A give $\kappa_{f,p} > 0$. Under kernel failure, scaling a direction in $\ker D_f$ with a positive policy loading gives objective value zero in (37).

Appendix C. Generated responses and weak-rank inference

C.1. Minimum-distance expansion

Let the minimum-distance criterion for family f be $L_{f,T}(\theta) = \hat{g}_{f,T}(\theta)' \widehat{W}_f \hat{g}_{f,T}(\theta)$. The approximate first-order condition and stochastic differentiability in Assumption 2 imply

$$0 = G_f' W_f \left\{ \hat{g}_{f,T}(\theta_f) + G_f(\hat{\theta}_f - \theta_f) \right\} + o_p(T^{-1/2})$$

uniformly on the maintained class. Multiplying by $\sqrt{T}$ and using nonsingularity of $G_f' W_f G_f$ yields

$$\sqrt{T}(\hat{\theta}_f - \theta_f) = -(G_f' W_f G_f)^{-1} G_f' W_f \sqrt{T} \hat{g}_{f,T}(\theta_f) + o_p(1).$$

Substitution of the moment influence representation proves (39). Uniform consistency keeps the expansion in the interior neighborhood on which the derivative and eigenvalue bounds hold.

C.2. Implicit equilibrium and response derivatives

At a selected equilibrium, total differentiation of $F_f\{z_f(\theta, q), \theta, q\} = 0$ gives

$$D_\theta z_f = -F_{f,z}^{-1} F_{f,\theta}, \quad D_q z_f = -F_{f,z}^{-1} F_{f,q}.$$

Applying the chain rule to $R_f(\theta, q) = \Phi_f\{z_f(\theta, q), \theta, q\}$ gives (41). A joint delta method applied to $(\hat{\theta}_f, \hat{q})$ then produces the first two terms in (40). If a simulated response is an average over S_T centered draws, its stochastic order is $S_T^{-1/2}$. After multiplication by $\sqrt{T}$, its limit scale is $\sqrt{T/S_T} \rightarrow \sqrt{\eta}$. Solver and discretization errors vanish from the first-order law by Assumption 2.

When two families use a common estimated treatment path or moments from overlapping samples, the associated influence vectors share components. Stacking before covariance estimation retains terms such as

$$R_{f,q} \mathbb{E}(\psi_q \psi_q') R_{g,q}' \quad \text{and} \quad R_{f,\theta} \mathbb{E}(\psi_{\theta_f} \psi_q') R_{g,q}'.$$

These terms enter the joint covariance through shared inputs and overlapping samples.

PROOF OF THEOREM 5. The minimum-distance expansion proves (39). Invertibility of $F_{f,z}$ gives the implicit derivatives. The joint functional delta method, centered simulation representation, and rate conditions then give (40). Stacking the primitive influence vectors and applying fixed linear response derivatives gives the joint family representation and its off-diagonal covariance blocks. $\square$

C.3. Uniform projection inference

PROOF OF THEOREM 6. For each null law P , choose a compatible nuisance ζ_P^0 with $g_{S,P}(\zeta_P^0, b_P) = 0$. By definition of the infimum,

$$P_{S,T}(b_P) \leq Q_{S,T}(\zeta_P^0, b_P).$$

Assumption 3 implies

$$\hat{\Omega}_S(\zeta_P^0)^{-1/2} \sqrt{T} \hat{g}_S(\zeta_P^0, b_P) \Rightarrow N(0, I_{d_S})$$

uniformly over the null class. Consequently the upper tail of $Q_{S,T}(\zeta_P^0, b_P)$ at $\chi_{d_S, 1-\alpha}^2$ converges uniformly to at most α . Since

$$\{P_{S,T}(b_P) > c\} \subseteq \{Q_{S,T}(\zeta_P^0, b_P) > c\},$$

taking the supremum over P and the limit superior proves (45). $\square$

The domination uses the value of g_S at a compatible model nuisance. Its derivative rank may change with P or T . Compactness and measurability make the profiled statistic measurable and support numerical approximation of the infimum. The optimizer returns an upper bound from evaluated feasible points and a convergence certificate; ϵ_{num} includes the remaining optimization gap.

C.4. Regular quotient refinement

PROOF OF THEOREM 7. Under constant rank, local coordinates (x, z) can be chosen so that the population moment factors through x . Assumption 4 gives the same factorization for the first-order covariance. After whitening at the truth and writing $u = \sqrt{T}(x - x_0)$, the uniform local quadratic expansion is

$$Q_{S,T}(x, z, b_0) = \|Z_T - Ju\|^2 + o_p(1), \quad Z_T \Rightarrow N(0, I_{d_S}),$$

where J has column rank r_S . Consistency of the local minimizer permits profiling on an $O_p(1)$ neighborhood. Therefore

$$P_{S,T}^{\text{reg}}(b_0) = \inf_u \|Z_T - Ju\|^2 + o_p(1) = \|(I - \Pi_J)Z_T\|^2 + o_p(1),$$

where $\Pi_J = J(J'J)^{-1}J'$ is the orthogonal projection onto $\text{Im } J$. The matrix $I - \Pi_J$ is idempotent with rank $d_S - r_S$. The continuous mapping theorem gives the limit in (46). $\square$

An active inequality boundary gives a local cone and a chi-bar-square or another boundary law. Covariance variation along z carries the fiber coordinate into the first-order experiment. The quotient refinement applies to interior cells with covariance factorization and locally constant rank.

C.5. Target-constrained quotient

PROOF OF THEOREM 8. Assumption 5 supplies local coordinates u on the target level set and a uniform expansion

$$Q_{A,f,T}(u, b_0) = \|Z_T - J_{A|b,f}u\|^2 + o_p(1), \quad Z_T \Rightarrow N(0, I_{d_A}).$$

Consistency permits profiling on this chart. The profiled criterion equals $\|(I - \Pi_{J_{A|b,f}})Z_T\|^2 + o_p(1)$. The projection matrix has rank $r_{f,b}$, which gives (49). A weak direction normal to the target level set is absent from $B_{f,b}$ and therefore absent from the profiled tangent score. $\square$

Appendix D. Application details and implementation

D.1. Data construction and units

The baseline quarterly sample is 1969Q1–2006Q4. The aggregate series are real output per capita (FRED A939RX0Q048SBEA), the GDP deflator (FRED GDPDEF), and the federal funds rate (FRED FEDFUNDS). Monetary shocks use the Romer and Romer (2004) series distributed with the Ramey data and the natural-language shock of Aruoba and Drechsel (2024). Monthly shocks are aggregated to quarters. The VAR uses two lags, an intercept, a linear trend, and the ordering AD shock, output, inflation, RR shock, and policy rate. Output and inflation enter horizons zero through twelve, and treatment paths are normalized to a 25-basis-point annualized policy-rate peak over horizons zero through twenty-four.

Under this design, $s_A^\dagger$ has 52 coordinates, the joint response and policy-rate input has 78 coordinates, and each delayed policy response $v_f(h; \theta)$ has 26 coordinates. The local Jacobian $D_{A,f}$ enters tangent calculations, while finite response secants enter the evaluated-design calculations. Inflation and the policy rate are divided by four at the model interface, and persistent interventions use the coefficients ρ^h applied to the delayed response vectors.

D.2. Nonlinear envelope computation

For each family and detection radius, the computation evaluates the equilibrium and the training and policy responses over the declared parameter domain, searches for extrema of training distance and policy targets from multiple starting points, and repeats the calculation under tighter solver tolerances and longer horizons. Family-specific contributions are then combined into the policy image of the compatible set. The numerical comparisons reported below use the retained baseline evaluated support for finite-support frontiers and outward-rounded boxes around selected parameter points for local continuum statements.

At the benchmark radius 3.9199, the retained-design family counts are 613, 561, 664, and 619. The combined counts across the eight testing-risk radii in equation (53) are 0, 12, 273, 1204, 2457, 3399, 3840, 3988. The first retained point with a nonnegative delayed or persistent target has distance 4.356667. A local HANK–RE search gives an inflation reversal witness at distance 4.253202.

D.3. Evidence subsets and the CMR expansion

The evidence exercise evaluates 26 response specifications on the same 4,000 structural points. The full-evidence specification reproduces the baseline distances, and the nested horizon and block-deletion results are reported in Tables A1 and A2.

TABLE A1. Training-horizon frontier on fixed support

Evidence	Retained	$c_{\text{entry}}^{\mathcal{D}}$	$c_{\text{sign}}^{\mathcal{D}}$	$Y < 0$	$\pi < 0$
Joint horizons 0–0	4,000	0.5962	0.5962	17/17	5/17
Joint horizons 0–1	4,000	1.6553	1.6505	17/17	5/17
Joint horizons 0–2	3,996	2.4831	2.4831	17/17	5/17
Joint horizons 0–3	3,884	3.1750	3.5993	17/17	5/17
Joint horizons 0–4	3,724	3.2940	3.6160	17/17	5/17
Joint horizons 0–5	3,653	3.3131	3.7090	17/17	5/17
Joint horizons 0–6	3,497	3.3459	3.8719	17/17	5/17
Joint horizons 0–7	3,396	3.3529	3.9460	17/17	17/17
Joint horizons 0–8	3,268	3.3646	4.0214	17/17	17/17
Joint horizons 0–9	3,078	3.3845	4.1165	17/17	17/17
Joint horizons 0–10	2,841	3.4391	4.2201	17/17	17/17
Joint horizons 0–11	2,588	3.4951	4.3080	17/17	17/17
Joint horizons 0–12	2,457	3.5058	4.3567	17/17	17/17

The support contains the same 4,000 baseline evaluations in every row. Retained counts and negative-cell counts use $c = 3.9199$, $\beta = 0.99$, and delays zero through sixteen. Each distance uses the principal covariance submatrix of its evidence block. $c_{\text{sign}}^{\mathcal{D}}$ is the first evaluated reversal distance; $c_{\text{entry}}^{\mathcal{D}}$ is the largest family-specific minimum distance. The parameter support, fitted shock inputs, and policy targets remain fixed.

TABLE A2. Shock–outcome blocks and deletion experiments

Evidence	Retained	$c_{\text{entry}}^{\mathcal{D}}$	$c_{\text{sign}}^{\mathcal{D}}$	$Y < 0$	$\pi < 0$
RR output	4,000	2.3584	2.3981	17/17	5/17
RR inflation	4,000	2.0878	3.3564	17/17	5/17
AD output	4,000	0.7816	0.7580	17/17	5/17
AD inflation	4,000	1.1097	1.4569	17/17	5/17
All except RR output	3,999	2.6609	3.5921	17/17	5/17
All except RR inflation	3,987	2.8592	3.0165	17/17	5/17
All except AD output	3,354	3.3685	4.2558	17/17	17/17
All except AD inflation	3,428	3.2548	4.2340	17/17	17/17

The support contains the same 4,000 baseline evaluations in every row. Retained counts and negative-cell counts use $c = 3.9199$, $\beta = 0.99$, and delays zero through sixteen. Each distance uses the principal covariance submatrix of its evidence block. $c_{\text{sign}}^{\mathcal{D}}$ is the first evaluated reversal distance; $c_{\text{entry}}^{\mathcal{D}}$ is the largest family-specific minimum distance. The parameter support, fitted shock inputs, and policy targets remain fixed.

The CMR calculation preserves the authors’ economic equations and calibration while adding 24 dated news states for the common announced-wedge experiment. The original and augmented contemporaneous responses agree within 6.54×10^{-12} . The empirical

fit has rank 50 and condition number 9.927. The paper reports this calibration as one evaluated extension point.

D.4. Local parameter neighborhoods

The convex response completion uses common simplex weights for each evaluated empirical response and its policy target. At the benchmark, its 2,457 vertices have maximum distance 3.919872 and joint sign margin 0.002193.

An outward-rounded 300-quarter calculation centers a positive-width box at each family’s minimum-distance retained point and holds the center-fitted shock path fixed within the box. Normalized halfwidths are 1.2×10^{-14} , 5.0×10^{-14} , 3.0×10^{-11} , and 2.0×10^{-13} for RANK-RE, HANK-RE, RANK-CD, and HANK-CD. The corresponding response-distance upper bounds are 3.5580, 3.5809, 3.7515, and 3.5788, and the largest target upper bounds are -0.006512 , -0.006703 , -0.010506 , and -0.010761 . These four boxes remain inside the benchmark and preserve negative signs in all 136 family–target cells.

TABLE A3. Outward-rounded finite-horizon native target subboxes

Family	Baseline distance	Halfwidth/span	Norm. volume	q	Largest upper	Port gap
RANK-RE	3.71518	1.0×10^{-12}	3.20×10^{-59}	0.786974	-0.006681	7.21×10^{-10}
HANK-RE	3.63938	5.0×10^{-13}	1.00×10^{-60}	0.797103	-0.006022	7.28×10^{-11}
RANK-CD	3.74878	1.0×10^{-11}	7.19×10^{-65}	0.739823	-0.005319	3.71×10^{-11}
HANK-CD	3.69479	1.5×10^{-11}	8.01×10^{-64}	0.823017	-0.005510	8.96×10^{-11}

Halfwidth is expressed as a fraction of each active coordinate’s retained-draw span; active volume is the product of the resulting normalized coordinate widths. The Neumann gate $q < 1$ certifies equilibrium nonsingularity throughout each box in the 300-horizon independent Python implementation. The maximum among 34 target upper bounds is negative in every row. Port gap is the largest absolute difference between the baseline target and the independent point implementation. The historical whole-box compatibility calculation used an interleaved input-column assembly. The corrected minimum-distance-box run reports 0 of 4 compatible boxes; its results are described in Appendix E. HANK household-grid error and the finite-horizon tail retain separate gates.

Horizon checks solve the margin-setting vertices at horizons 200, 225, 250, 275, and 300. At the binding RANK-CD inflation horizon-sixteen cell, the $T = 200 - T = 300$ difference is 7.843×10^{-7} , or 0.03577 percent of the 0.0021925988 margin. Across the 2,457 compatible vertices, the maximum $T = 200 - T = 300$ response discrepancy is 2.682×10^{-5} and the maximum relative equilibrium residual is 2.235×10^{-16} .

TABLE A4. Target-unit checks at the binding sign-margin cell

Quantity	Target units	Margin share	Scope
Artifact- $T = 300$ difference	2.987×10^{-10}	0.0000136%	Binding vertex
$T = 200 - T = 300$ difference	7.843×10^{-7}	0.03577%	Fixed-grid vertex
$T = 275 - T = 300$ difference	3.728×10^{-8}	0.001700%	Fixed-grid vertex
Local policy secant discrepancy	0.001341	61.16%	Observed vertex secant
Factor-relation secant discrepancy	0.001418	64.69%	Observed vertex secant
All-vertex artifact- $T = 300$ maximum	2.375×10^{-8}	0.001083%	83,538 comparisons
All-vertex $T = 200 - T = 300$ maximum	2.682×10^{-5}	1.223%	83,538 comparisons
All-vertex local-policy secant maximum	0.3424	15,616%	Baseline design
All-vertex factor-relation secant maximum	0.1658	7,561%	Baseline design
Conditional standard error	0.015442	704.28%	Local influence map
Evaluated 95% projection displacement	0.002716	123.89%	Union upper endpoint

The reference margin is 0.0021925988 at RANK-CD inflation horizon sixteen. Fixed-grid rows use independent equilibrium solves. All-vertex rows cover the 2,457 benchmark-compatible evaluations and 34 targets. Secant maxima measure approximation error on the baseline design. The conditional standard error uses the empirical-response component of the generated-response influence stack.

Four response-center and covariance specifications provide a sensitivity comparison. The baseline admits 2,457 evaluations with margin 0.002193; the current-vintage center admits 2,417 with the same margin; the model-response training covariance admits 3,215 with margin 0.001470; and the rank-50 pseudoinverse yields all-family entry radius 4.2862. The first evaluated reversal radii are 4.3567, 4.3471, 4.2391, and 5.0199 in the same order.

TABLE A5. Baseline-design response-center and covariance sensitivity

Specification	Rank	Count	Margin	Coverage	Reversal
Baseline, tapered	52	2,457	0.002193	3.5058	4.3567
Current-vintage center, tapered	52	2,417	0.002193	3.5125	4.3471
Baseline, raw pseudoinverse	50	0	n/a	4.2862	5.0199
Model-artifact training-only	52	3,215	0.001470	3.3916	4.2391

Compatible counts use radius 3.9199. Family-entry radius is the largest family minimum over the baseline design. Reversal radius is the first evaluated delayed-target sign reversal. Each populated row and its joint response-policy convex completion retains a negative upper endpoint. The raw rank-50 covariance gives an empty benchmark set.

D.5. Weak-rank simulation design

The local model dimension is six and sample size is $T \in \{150, 300, 600\}$. Let D_T and C_T be the empirical-response and scalar-target Jacobians. The smallest empirical singular value follows

$$(A1) \quad \sigma_{\min}(D_T) = \frac{c}{\sqrt{T}}, \quad c \in \{0, 0.25, 0.5, 1, 2, 5\},$$

and the target loading on the associated right singular vector n_T is

$$(A2) \quad C_T n_T = \gamma, \quad \gamma \in \{0, 0.25, 0.5, 1\}.$$

Generated-response uncertainty varies the correlation between the response and Jacobian influence projections over $\{0, 0.5, 0.9\}$ and their standard-deviation ratio over $\{0, 0.5, 1\}$. Each cell uses 5,000 replications.

Among the cells with $c \leq 1$ and $\gamma > 0$, projection has mean rejection 0.146 percent and mean unbounded-set share 74.318 percent. The target-constrained quotient has mean rejection 4.996 percent and mean unbounded-set share 67.306 percent. In the fixed-positive-singular-value cells, regular quotient rejection ranges from 4.38 to 5.74 percent and the unbounded-set share is zero. These calculations illustrate the behavior of projection and quotient procedures as the response Jacobian approaches weak rank.

D.6. Output-stabilization diagnostic

The output-stabilization exercise sets $T = 13$, $A_Y = I_{13}$, $A_\pi = A_i = 0$, and $\wedge = 0$. For each compatible family draw, the first thirteen delayed output columns form Y_m , and the rule solves $m = -Y_m^{-1} y_x$ before carrying the same coefficients into inflation.

TABLE A6. Historical output-stabilization rule on the compatible retained design

Family	Draws	$\max r_Y $	Median policy peak	Maximum policy peak	Inflation range
RANK-RE	613	2.66×10^{-15}	115.48	218.45	[42.16, 78.71]
HANK-RE	561	1.33×10^{-15}	42.68	101.66	[42.32, 78.54]
RANK-CD	664	2.66×10^{-15}	116.41	208.62	[40.14, 57.47]
HANK-CD	619	1.55×10^{-15}	39.95	86.99	[40.48, 65.30]

r_Y is the 13-quarter output residual after solving the output-stabilization rule for the fixed 2021Q2 OLS forecast. Policy peaks are annualized percentage points. Inflation range reports the $\beta = 0.99$ discounted cumulative annualized inflation path; its baseline value is 38.15. Every 13×13 output-response matrix has rank 13. Direct and normal-equation solutions differ by at most 1.20×10^{-8} .

Every output-response matrix has rank 13 and the maximum direct-solve residual is 2.66×10^{-15} . Exact output stabilization raises discounted inflation from 38.15 to 40.14–78.71 and requires policy adjustments with family medians 39.95–116.41 annualized percentage points; the maximum is 218.45. These values describe the exact-stabilization criterion in the historical scenario.

D.7. Local-projection response estimator and structural profiling

For raw quarter t , let w_t multiply its Hamilton-regression contribution and its VAR residual or LP forecast-origin contribution. The estimator $\hat{s}(w)$ refits both stages and peak normalization. With $D_t = \partial \hat{s}(w) / \partial w_t|_{w=1}$ and bandwidth b , the finite-sample covariance is

$$(A3) \quad \widehat{\Sigma}_b = \sum_t D_t D_t' + \sum_{\ell=1}^b \left(1 - \frac{\ell}{b+1}\right) \sum_{t=\ell+1}^{275} (D_t D_{t-\ell}' + D_{t-\ell} D_t').$$

The LP impact matrix is the Cholesky factor from a VAR with three lags. Direct horizon regressions use current variables, three lags, an intercept, and a trend. Forecast-origin samples shrink with horizon. The main refitting calculation uses bandwidth 28 and reports bandwidths 8 and 16 as sensitivity rows.

For fixed structural parameters, fifty shock coefficients are represented through fifty normalized policy-rate ordinates. Two peak ordinates and two recursive-zero equalities impose the normalization restrictions. After whitening and eliminating the fixed peaks, the inner problem is a convex least-squares criterion on an affine box. A linear program computes a first-order gap bound for each fitted point. The structural search evaluates 102 distinct parameter points across the four families and reprofiles each point under both coordinate choices and all three bandwidths, giving 612 constrained fits on a common support.

TABLE A7. LP-refitted radius pairs across HAC bandwidths

Experiment	Bandwidth	Family entry	First reversal	Count at 3.9199
Training 50	8	14.3999	11.1563	0
Training 50	16	17.5390	14.2231	0
Training 50	28	23.2840	17.1449	0
Full 146	8	204.0180	150.8373	0
Full 146	16	252.6573	211.4665	0
Full 146	28	323.1614	290.0486	0

All six rows use the same 102 evaluated parameter points. Shock paths are reprofiled for each experiment and covariance. The numerical search generated these points under the declared objective and parameter hull; every reported distance is reevaluated at 300 quarters. In every row, reversal precedes all-family entry.

Across the common-support profiles, the maximum normalization equality error is 3.56×10^{-15} , the maximum rate-bound excess is 2.64×10^{-15} , and the largest relative inner-profile gap is 3.74×10^{-10} . The selected entry and reversal points change by at most 0.000126 in distance between 250- and 300-quarter solves, and the corresponding target change is at most 3.59×10^{-6} .

TABLE A8. Horizon checks at LP entry and reversal evaluations

Family	Point	Distance at 300	$ d_{250} - d_{300} $	Largest target change
RANK-RE	entry	23.2840	1.57e-08	2.01e-09
RANK-RE	first reversal	32.2795	2.06e-07	1.50e-09
HANK-RE	entry	18.2001	3.63e-07	1.41e-06
HANK-RE	first reversal	25.8588	1.26e-04	5.77e-07
RANK-CD	entry	22.3531	9.51e-06	3.59e-06
RANK-CD	first reversal	27.1542	4.54e-09	3.45e-10
HANK-CD	entry	17.0400	1.25e-08	1.63e-08
HANK-CD	first reversal	17.1449	1.54e-06	1.24e-06

Distances use the 50-coordinate LP experiment and bandwidth 28; shock paths are reprofiled at each horizon. Each first-reversal evaluation has a positive target at both horizons. These pointwise comparisons keep the baseline HANK household grid fixed.

D.8. Experiment-gap sampling calculations

For a profiled distance $d_i = \|L^{-1}(s_i - \hat{s})\|$, the envelope derivative with respect to the empirical center is

$$(A4) \quad g_i = -L^{-1} \frac{L^{-1}(s_i - \hat{s})}{d_i}.$$

Gaussian draws update each distance by $d_i^{(b)} = d_i + g_i' Z^{(b)}$ and recompute family minima, all-family entry, and the first sign crossing across the 102 evaluated points. The HAC-8, HAC-16, and HAC-28 upper endpoints of the 95-percent gap ranges are -1.2974 , -1.3484 , and -4.2955 .

For the common-support decomposition, the 102 LP-profiled fitted responses and policy targets remain fixed while response dynamics, peak normalization, and covariance are changed in sequence. Under bandwidth 28, the gaps are -0.0683 , -0.9548 , -1.0918 , and -6.1391 . A second decomposition compares the 4,000 baseline points, the common 102-point parameter support under the original shock fit, and the same support under LP shock profiling. Their HAC-28 gaps are 3.3811 , 6.2580 , and -6.1391 .

The initial calendar block-multiplier calculation reruns Hamilton detrending, lag-augmented LP estimation, impact identification, and response construction while updating profiled distances through local envelope derivatives. Under adaptive peak selection, block lengths 8, 16, and 28 give 95-percent ranges $[-5.8329, 0.4248]$, $[-6.4171, 1.0715]$, and $[-8.5502, 0.9632]$. The fixed-multiplier and fixed-peak variants also produce ranges spanning zero. Table A9 records the full grid examined before the selected full-reprofile calculation.

TABLE A9. Calendar-bootstrap specification grid preceding full reprofiling

Block	Normalization	95% gap range	Pr(gap > 0) (%)	Peak switches
8	Fixed multiplier	$[-5.6159, 0.3616]$	5.503	0
8	Fixed peak cell	$[-6.2591, 0.5773]$	5.403	0
8	Adaptive peak	$[-5.8329, 0.4248]$	4.702	1,458
16	Fixed multiplier	$[-5.5025, 1.0150]$	14.907	0
16	Fixed peak cell	$[-6.9476, 1.1636]$	8.854	0
16	Adaptive peak	$[-6.4171, 1.0715]$	7.354	1,433
28	Fixed multiplier	$[-7.9836, 0.7461]$	5.253	0
28	Fixed peak cell	$[-8.7704, 1.2314]$	4.752	0
28	Adaptive peak	$[-8.5502, 0.9632]$	4.302	1,431

Every row uses 1,999 circular calendar block-multiplier draws on the 275-quarter raw-data vintage. Each draw reruns response estimation and then updates the profiled distances through local envelope derivatives on the 102-point support. The HAC metric, structural support, policy targets, and active profile derivative remain fixed. This grid records the bandwidth and normalization specifications examined before full shock-path reprofiling.

The selected HAC-28 calculation recomputes every constrained shock path and both frontier binders in each of 1,999 draws. It gives a point gap of -6.1391 , bootstrap quantiles $(-10.2187, -4.9865, 2.2214)$ at probabilities $(0.025, 0.5, 0.975)$, and a 6.353-percent positive-gap frequency. All draws produce a frontier. Across 203,898 point profiles, 240 constraint systems are infeasible and the remaining profiles determine every reported frontier. The maximum equality error is 2.49×10^{-14} , the maximum rate-bound excess is 9.92×10^{-15} , and the maximum propagated gap-envelope width is 3.01×10^{-5} .

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